Project Guidelines

Project description

Objective (what we are trying to achieve).

The goal of this project is to apply optimization techniques to solve a real-world data science problem of your choice, demonstrating their effectiveness in practical applications in a group of 2. It is also possible to do individual project on your own.

Scope (what does the project include and exclude).

Students will focus on an optimization problem relevant to data science, formulate an appropriate mathematical model, implement various solution methods, and analyze their accuracy, efficiency, and interpretability.

Project Idea Presentation

Each team will deliver a 10-minute in-class presentation of their proposed project for feedback during Week 6 Class 2. The presentation should briefly introduce the group members, describe the problem to be addressed, outline potential data sources, and summarize the planned methodology.

Report Format.

The report should be no more than 10 pages, which corresponds to 5 double-sided pages, excluding the cover page, appendix, code, datasets, and references. Each team will present the problem and their main findings in a 10-minute in-class presentation during Week 13 Class 2, using approximately 5–6 slides.

Template for the Report

- **Problem Identification**
  - Clearly describe the problem in plain English.
- **Problem Modeling**
  - Formulate the problem as an optimization problem.
  - Clearly define decision variables, constants, constraints, and the objective function.
  - Specify the type of optimization problem (e.g., linear, convex, nonconvex).
  - State any assumptions made in the modeling process.
- **Model Correctness**
  - Justify why the formulated model accurately represents the problem.
- **Solving Problem Instances Using Software**
  1. Choose an appropriate optimization approach and implement the solution using gradient-based algorithms in Python (e.g., in Google Colab or Jupyter Notebook) to solve instances of the problem.
  2. Use real (if available) or synthetic data.

3. Include well-commented code in the appendix.
4. Analyze and interpret results, discussing computational efficiency and solution quality.

A concrete example

Problem Identification.

The goal of the project is to predict house prices based on historical data containing house features. This is relevant in real estate pricing, finance, and other domains.

Problem Modeling.

We apply optimization techniques to a linear regression problem, where the goal is to predict house prices based on features such as square footage, number of bedrooms, and location. We formulate linear regression as an optimization problem:

- **Decision Variables:** Let x_i be the feature vector for house i, where $x_i \in \mathbb{R}^n$ (e.g., square footage, number of bedrooms, etc.). Let y_i be the true price of house i. Let $\beta \in \mathbb{R}^n$ be the vector of regression coefficients. Let β_0 be the intercept term.
- **Objective Function:** We minimize the sum of squared residuals:

$$\min_{\beta,\beta_0} \sum_{i=1}^{m} \left(y_i - (\beta_0 + x_i^\top \beta) \right)^2.$$

- **Constraints:** No strict constraints in standard linear regression.

We also consider the effect of regularization:

- **Lasso Regression (ℓ_1 regularization):**

$$\min_{\beta,\beta_0} \sum_{i=1}^{m} \left(y_i - (\beta_0 + x_i^\top \beta) \right)^2 + \lambda \sum_{j=1}^{n} |\beta_j|.$$

- **Ridge Regression (ℓ_2 regularization):**

$$\min_{\beta,\beta_0} \sum_{i=1}^{m} \left(y_i - (\beta_0 + x_i^\top \beta) \right)^2 + \lambda \sum_{j=1}^{n} \frac{\beta_j^2}{2}.$$

Model Correctness.

The formulated model correctly represents the problem because:

- The least squares formulation minimizes prediction error.
- Regularization prevents overfitting.
- The feature vector x_i and output y_i match the assumptions of a linear model.

Solving Instances Using Software.

We implement and solve the problem in Python, utilizing either synthetic data or real data from publicly available datasets. The problem is solved using vanilla least squares regression as well as regularized regression with ℓ_1 (Lasso) and ℓ_2 (Ridge) regularization. We report a comparison of performance metrics (e.g., mean squared error) and provide insights into accuracy, computational efficiency, and interpretability of the various models.

Project suggestions

Some good project suggestions include:
- Support vector machines (e.g., for handling data that is not linearly separable).
- Kernel methods (applying SVMs in a higher-dimensional feature space).
- Logistic regression (e.g., Softmax regression, a generalization of logistic regression to the case where we want to handle multiple classes).
- Regularized linear regression (e.g., penalizing residuals or model parameters).
- Markowitz portfolio optimization (e.g., use Conditional Value at Risk (CVaR)).
- Principal component analysis.
- Nonnegative matrix factorization.

An additional valuable resource for project ideas and optimization models is our Convex Optimization textbook. In particular, the Applications section provides over 150 pages of diverse models across various domains.

Grading rubrics

Assessment will be based on the following:
- Correctness, clarity, and quality of your report,
- Appropriate level of working and explanation provided (including in any code files),
- Appropriate, sensible, and even creative use of course material (you may use any part of the course),
- Presentation and proper citation of all sources.

Timeline

- **Week 3:** Project released.
- **Week 6 Class 2:**
  Present your project idea for 10 minutes for feedback.
  The presentation should include:
  - **Names and student IDs of 2 group members (1 if you are doing an individual project)**
  - A brief description of the problem,
  - Potential data source(s),
  - Methodology to use.
- **Week 13 Class 2:** Project presentation, 10 minutes for group.
- **Monday Week 14,** 10 pm:
  Final project submission due, including:
  - Project report (maximum 10 pages),
  - Presentation deck,
  - Excel sheets with data for the problem,
  - Codes with documentation.